Effect of Anion Receptor Additives on Electrochemical Performance of Lithium-Ion Batteries

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Four boron-based anion receptors were investigated as electrolyte additives for lithium-ion batteries. The electrochemical performance of lithium-ion cells was found to strongly depend on the structure of the anion receptor added to the electrolyte. The capacity retention of the lithium-ion cell was slightly improved by adding 0.07 M bis(1,1,1,3,3,3-hexafluoroisopropyl)pentafluorophenylboronate additive, whereas the addition of 2,5-bis(trifluoromethylphenyl)tetrafluoro-1,3,2-benzodioxaborole dramatically deteriorated the electrochemical performance. The addition of a certain type of anion receptor can promote the electrochemical decomposition of the electrolyte, resulting in high interfacial impedance and accelerated capacity fading of lithium-ion cells. *Ab initio* calculations showed that the electrochemical performance of anion receptors had good correlation to the degree of localization of the lowest unoccupied molecular orbital at the boron center of anion receptors, which can potentially be used in the search for new anion receptors for lithium-ion batteries.

Introduction

Anion receptors are a class of organic ligand that can efficiently and selectively coordinate with anions and negatively charged functional groups like carboxylate and phosphate through $\pi-\pi$ coordination bonds or hydrogen bonds.[1−8] The fundamental research on this subject, anion coordination chemistry, can be traced back to the 1960s, during which Simmons and co-workers reported that macrobicyclic amines with the proper cavity diameter can efficiently coordinate with halide anions.[1] Simmons's report immediately triggered increasing interest in exploring new anion receptors for higher efficiency and selectivity.[2−4] After decades of research in this area, anion receptors have successfully gained important applications in chemical and biochemical systems for anion coordination, recognition, and transportation[9,10] as well as nonaqueous electrolytes for lithium batteries.[11−34]

Yang and co-workers reported a series of anion receptors that are strong Lewis acids, forming strong $\pi-\pi$ or donor−acceptor interactions with anions in nonaqueous electrolytes.[11−13,15,17,19,24] The formation of ion pairs in electrolytes is one of the key factors limiting the ionic conductivity of the electrolytes. When the anion receptor is added to the electrolyte, a cluster of anion receptors and anions is formed, and part of the negative charge from the anion is delocalized to the anion receptor via $\pi-\pi$ interaction, leading to decreased charge density on the anion and less electric field attraction between cations (Li^+) and anions. Therefore, the addition of an anion receptor can lead to higher cation mobility and higher ionic conductivity.[11,23] Some of these anion receptors are strong enough to break up the interaction between the Li^+ cation and F^- anion and can help to dissolve LiF,[16,18,28] Li_2O, and Li_2O_2[29] in nonaqueous solvent. Inspired by these studies, Chen and co-workers proposed to use a low concentration of anion receptors as an electrolyte additive to dissolve inorganic components like LiF and Li_2CO_3 in the solid−electrolyte interphase (SEI), which, in turn, would reduce the interfacial impedance[25] and improve the capacity retention of lithium-ion cells.[18,25,26,30] Excess anion receptor in nonaqueous electrolytes can also accelerate the decomposition of electrolytes, leading to thickening SEI layer.[20,22,25,33] Although this effect may lead to better protection of the carbon-based negative electrode and better capacity retention of the lithium-ion cells,[18,25,26,30] it can also lead to higher impedance, resulting in power loss of lithium-ion batteries,[25] which is a serious problem for automobile applications. Therefore, an improved formulation is needed to maximize the functionality of the anion receptor additive and minimize its negative effect on the power capability.

As part of an effort to establish the structure−properties relationship of anion receptors used as the battery electrolyte additive, Chen and co-workers applied density function theory (DFT) to estimate the fluoride affinity of anion receptors.[31] On the basis of the theoretical calculation, several boron-based compounds were identified to have higher fluoride affinity than the tris(pentafluorophenyl)borane (TPFPB),[31] which was previously reported as a promising additive to improve the power capability of lithium-ion batteries.[25] As a continuation of our previous theoretical effort, in this paper, we will report the results of our electrochemical characterization on four anion receptors with predicted strong fluoride affinity. We also used DFT to understand the performance discrepancies among the selected candidates. The results show that the additives with stronger fluoride affinity may not simply lead to better capacity retention of the lithium-ion cells. Many other factors, especially the increased impedance, resulted from the electrolyte decomposition are playing important roles.

Experimental Section

Figure 1 shows the molecular structures of the anion receptor candidates that were synthesized, as reported previously.[24] These candidates are 2-(2,4-difluorophenyl)tetrafluoro-1,3,2-benzodioxaborole (AR01), 2,5-bis(trifluoromethylphenyl)tetrafluoro-1,3,2-benzodioxaborole (AR02), tris(1,1,1,3,3,3-hexafluoroiso-

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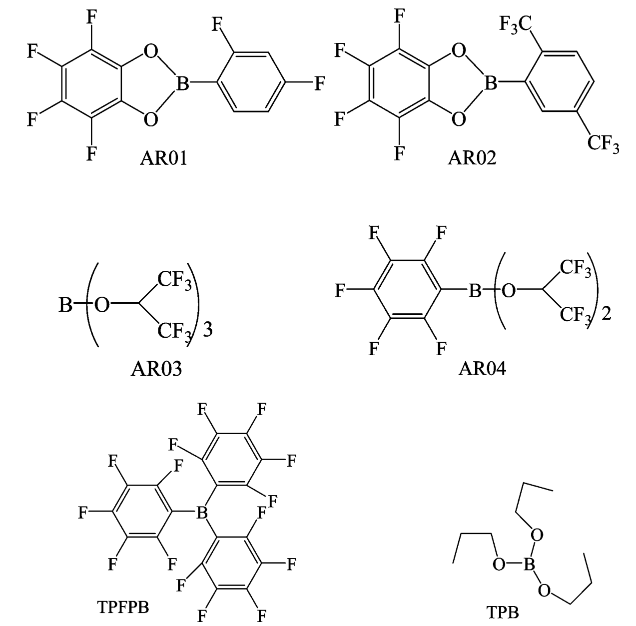

Figure 1. Molecular structure of investigated anion receptors. These candidates are 2-(2,4-difluorophenyl)-tetrafluoro-1,3,2-benzodioxaborole (AR01), 2,5-bis(trifluoromethylphenyl)-tetrafluoro-1,3,2-benzodioxaborole (AR02), tris(1,1,1,3,3,3-hexafluoroisopropyl) borate (AR03), bis(1,1,1,3,3,3-hexafluoroisopropyl)pentafluorophenylboronate (AR04), tris(pentafluorophenyl)borane (TPFPB), and tripropyl borate (TPB).

propyl) borate (AR03), and bis(1,1,1,3,3,3-hexafluoroisopropyl)pentafluorophenylboronate (AR04).

Figure 1 also shows the molecular structure of TPFPB and tripropyl borate (TPB), whose electrochemical data had been reported previously[25,34] and were further investigated by *ab initio* calculations in the present study.

The anion receptors AR01 to AR04 were tested in 2032-type coin cells. The positive electrode was made of 84 wt % $Li_{1.1}[Ni_{1/3}Co_{1.3}Mn_{1/3}]_{0.9}O_2$ (NCM), 8 wt % TB5500 carbon black, and 8 wt % poly(vinylidene fluoride) (PVDF, Kurela 7208). The negative electrode was made of 90 wt % mesocarbon microbeads (MCMB), 2 wt % vapor grown carbon fiber (VGCF), and 8 wt % PVDF (Kurela 1100).

Each coin cell was configured with a negative electrode (MCMB), a microporous polypropylene separator (Celgard 2325), a positive electrode (NCM), and an appropriate amount of electrolyte. The electrolyte was 1.2 M $LiPF_6$ dissolved in a mixture solvent of ethylene carbonate (EC) and ethylmethyl carbonate (EMC) with a ratio of 3:7 by volume, with or without anion receptor additive.

The cells were first subjected to three formation cycles at the C/10 rate and then cycled at the 1 C rate with voltage between 3.0 and 4.0 V. Before and after 200 cycles, the cells were constant voltage charged to 3.8 V and then taken out of the oven and cooled to room temperature for ac impedance investigation using a BAS-Zahner IM6 impedance analyzer (Zahner Electric). The frequency window was between 1 MHz and 0.01 Hz, and the amplitude of the stimulus signal was 5 mV.

All the calculations reported in this work were carried out with the Gaussian03 computational package.[35] Hartree−Fock theory with a basis set of 6-31G(d,p) [HF/6-31G(d)] was used for full geometry optimization and calculation of the zero-point energy correction and the thermal correction to the standard free energy at 298 K. The single-point electric energy was determined from *ab initio* calculations using B3LYP, a well-known hybrid model of DFT with the Becke three-parameter function for electron exchange[36] and the Lee−Yang−Parr function for electron correlation.[37] The basis set for the single-point energy calculations is 6-31+G(d).

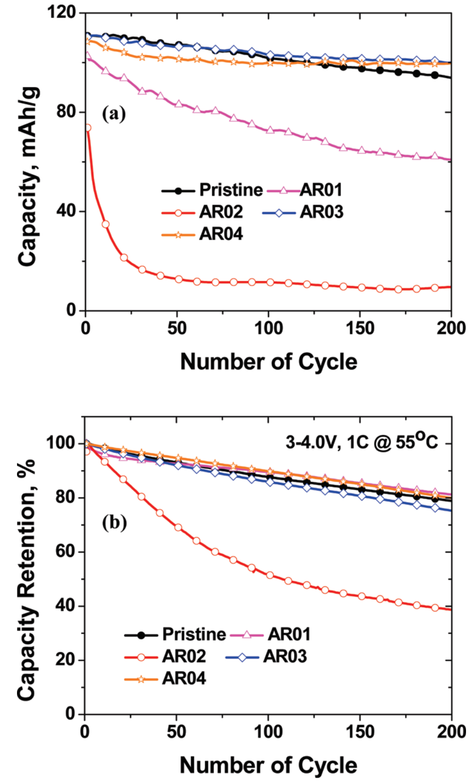

Figure 2. (a) Discharge capacity of MCMB/NCM cells with and without anion receptor additive cycled between 3 and 4.0 V at 1 C rate and room temperature. (b) Capacity retention of another set of cells charged/discharged at 55 °C.

Results and Discussion

Figure 2a shows the discharge capacity of MCMB/NCM lithium-ion cells that were cycled between 3.0 and 4.0 V with a constant current of 1 C (∼2 mA) at room temperature. The baseline electrolyte used was 1.2 M $LiPF_6$ in EC/EMC (3:7). For those cells using anion receptor additives, the concentration of the anion receptors were 1.8 wt % for AR01, 2.4 wt % for AR02, 3.0 wt % for AR03, and 3.0 wt % for AR04. The initial concentration was picked to maintain a constant molarity of 0.07 M/L, which is equivalent to the recommended concentration (1.0 wt % or 0.07 M/L) for the best performance[25] with TPFPB additive. Figure 2a shows that electrolyte using additives AR03 and AR04 only slightly improved the capacity retention. However, the capacity retention was greatly deteriorated by the additives AR01 and AR02. This observation is just opposite to the impact of TPFPB,[25] which was reported to enhance the capacity retention when used as an additive. A possible explanation of this discrepancy is that the anion receptors investigated here have even higher fluoride affinity than TPFPB,[31] thereby increasing the decomposition of $LiPF_6$ and resulting in higher interfacial impedance. Thus, another series of cells was prepared and cycled at 55 °C to confirm the negative effect of increased interfacial impedance (see Figure 2b). The cycling data showed that the cells with AR01, AR03, and AR04 had similar capacity retention to that of the cell without any additive. The cell with 2 wt % AR02 showed improvement in the reversible capacity and capacity retention compared with

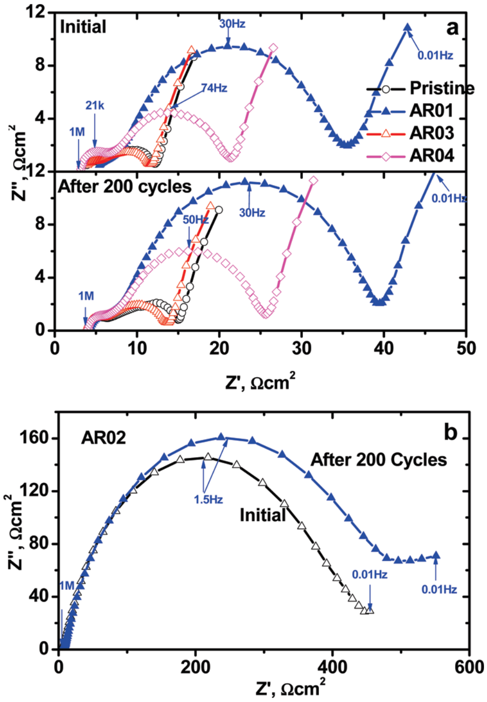

Figure 3. Ac impedance of MCMB/NCM cells with and without anion receptors before and after 200 cycles at room temperature.

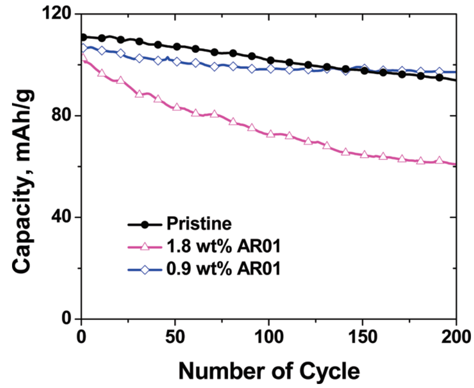

Figure 4. Discharge capacity of MCMB/NCM cells with and without AR01 additive cycled between 3 and 4.0 V at 1 C rate and room temperature.

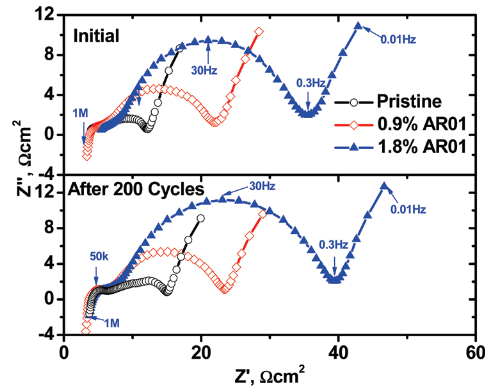

Figure 5. Ac impedance of MCMB/NCM cells with and without AR01 anion receptor before and after 200 cycles at room temperature.

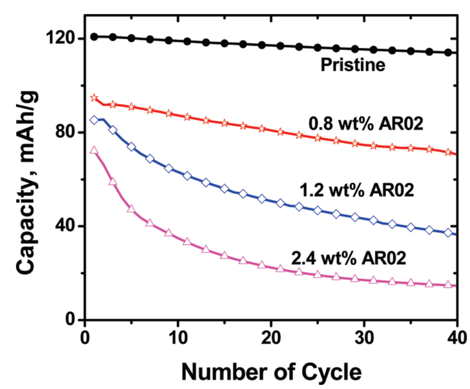

Figure 6. Discharge capacity of MCMB/NCM cells with different concentration of AR02 additive cycled between 3 and 4.0 V at 1 C rate and room temperature.

the room-temperature test, but its performance was still much worse than the cell without any additive.

Figure 3 shows the ac impedance of lithium-ion cells whose cycling data are shown in Figure 2a. The impedance data were collected after the cells were constant voltage charged to 3.8 V. Figure 3b shows that the impedance of the cell using the AR02 additive was significantly higher (>10 times) than that of the other cells. Figure 3a shows that the addition of 3 wt % AR03 slightly reduced the interfacial impedance, while the other three candidates significantly increased the impedance. The anion receptors mostly affect the low-frequency semicircle (in the range of tens of hertz) in the impedance spectra, which is related to the charge transfer reaction at the SEI. This finding agrees well with our hypothesis that the addition of excess amount of strong anion receptor to the electrolyte can lead to thickening of the SEI layer and increased interfacial impedance as well as the capacity retention shown in Figure 2a.

To investigate the effect on additive concentration, we conducted further tests with different concentrations of AR01 and AR02, which showed poor electrochemical performance (see Figure 2a). As the first trial, the AR01 concentration was reduced by half (from 1.8 to 0.9 wt %). Figure 4 shows the capacity retention of the lithium-ion cells with 0.9 wt % AR01, 1.8 wt % AR01, and no additive. The data show that the initial reversible capacity of the cells decreased with the concentration of AR01 added, probably due to the irreversible capacity loss caused by SEI formation. With the 0.9 wt % AR01, however, the capacity retention over 200 cycles was slightly better than that of the cell without additive. Figure 5 shows that reducing the AR01 concentration also cut the interfacial impedance growth proportionally. This finding agrees well with our previous report on other anion receptors that the addition of the proper amount of anion receptors to the electrolyte can improve the capacity retention of lithium-ion cells.[25,26,34]

Figure 6 shows the capacity retention of lithium-ion cells with different AR02 concentrations. When the concentration of AR02 decreased, the initial irreversible capacity of the cells increased, and the capacity retention improved. However, the performance of the 0.8 wt % AR02 cell was still much worse than that of the cell without additive.

From the above discussion, one can easily see a dramatic discrepancy in the electrochemical performance of lithium-ion cells using different anion receptors as electrolyte additive. The addition of 0.07 M AR01 or AR02 led to a significant impedance increase and accelerated capacity fade; the addition of 0.07 M AR04 led to a slight impedance increase and better capacity retention. It was previously reported that addition of TPFPB can both reduce the interfacial impedance and improve the capacity retention,[25] while the addition of TPB had little impact on both impedance and capacity retention.[34] Using impedance and capacity retention as the evaluation criteria, we would order the performance of anion receptors roughly as follows:

TABLE 1: Chemical Properties of Anion Receptors Calculated Using Density Functional Theory

	AR01	AR02	AR03	AR04	TPFPB	TPB
ΔG^{sol} (kJ/mol)[a] (F^- complexing)	−1255.0[b]	−1269.7[b]	−1161.4	−1310.6[b]	−1288.7[b]	−1163.8
B−F bond in B−F−P (Å)[c]	2.604	2.626	2.455	2.549	2.219	2.613
P−F bond in B−F−P (Å)[c]	1.628	1.631	1.645	1.645	1.665	1.624

[a] Continuous solvation model in acetonitrile was used to calculated the single point energy. [b] From ref 31. [c] The bond length was obtained from the optimized ground state of the AR−PF_6^- complex. The length of P−F bond in PF_6^- anion is 1.606 Å.

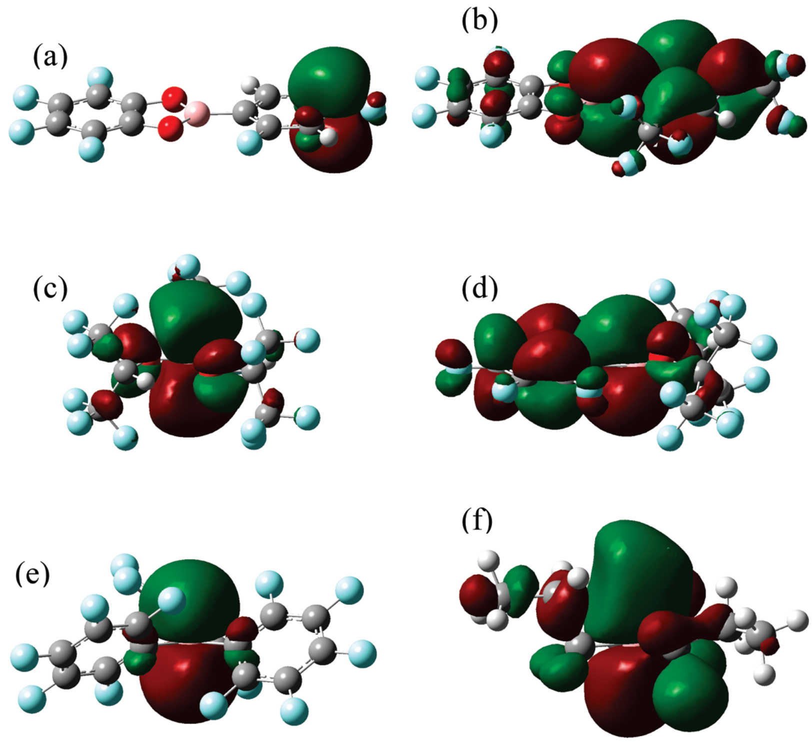

Figure 7. Schematic illustration of lowest unoccupied molecular orbital (LUMO) of (a) AR01, (b) AR02, (c) AR03, (d) AR04, (e) TPFPB, and (f) TPB calculated by GaussianW03.

$$\text{TPFPB} > \text{AR03} > \text{AR04} > \text{TPB} > \text{AR01} > \text{AR02} \quad (1)$$

These differences might originate from some fundamental properties of anion receptors that have not been taken into consideration. Therefore, we performed DFT calculations to further understand the structure−property relationship of the anion receptors (AR01−AR04, TPFPB, and TPB).

It was reported previously that a small amount of anion receptor can help to dissolve some inorganic components like LiF or Li_2O in the SEI layer, thereby improving the ionic conductivity of the SEI layer and the cycling performance of lithium-ion cells,[25,26,34] and that excess anion receptor can also accelerate the decomposition of $LiPF_6$ and increase the concentration of PF_5, which is a catalyst for accelerating the ring-opening polymerization of ethylene carbonate.[38] Hence, the strength or fluoride affinity of anion receptors can be a critical property to explain the significantly different behavior of anion receptors in lithium-ion cells.

We first calculated the change of free energy when the anion receptor combines with a fluorine anion. Table 1 shows that the calculated free energy change (ΔG^{sol}) ranges from −1161.4 kJ/mol (AR03) to −1310.6 kJ/mol (AR04). The above calculation is based on the two-step reaction:

$$Li^+ + PF_6^- \rightarrow PF_5 + LiF \qquad \Delta G_2^{sol} \quad (2)$$

$$LiF + AR \rightarrow Li^+ + AR{-}F^- \qquad \Delta G_3^{sol} \quad (3)$$

Apparently, a strong anion receptor with low ΔG^{sol} will enhance reaction 3 and indirectly increase the residue PF_5 concentration in the electrolyte, resulting in higher interfacial impedance. However, there is no clear correlation between the calculated ΔG^{sol} (see Table 1) and the electrochemical performance of lithium-ion cells containing anion receptors (see eq 1). An alternative explanation is that the decomposition reaction is controlled by the kinetics of the reaction instead of thermodynamics. It is well-known that LiF has extremely low solubility in the electrolyte, which can significantly limit the rate of reaction 3. Therefore, some anion receptors can take an alternative reaction path by coordinating with the PF_6^- anion first and then directly breaking the P−F bond to form PF_5. Our optimization effort to locate the transition state of the AR−PF_6^- cluster led to ground states with lower energy level. Table 1 lists the length of the B−F and P−F bonds in B−F−P for the optimization structure of the AR−PF_6^- cluster. The coordination of PF_6^- with anion receptors all extended the length of the P−F bond (compared with 1.606 Å for no additive). Unfortunately, the bond length of B−F and P−F does not correlate well with the electrochemical performance order of the additives shown in eq 1.

We also examined the difference in the electronic structures of anion receptors. Figure 7 shows the lowest unoccupied molecular orbital (LUMO) of the investigated anion receptors. The balloons on each atom represent the contribution of its atomic orbital to LUMO. When the anion receptor coordinates with an anion like PF_6^-, the lone electron pair on the anion will be partially donated to the LUMO of the anion receptor via $\pi-\pi$ interaction. The LUMO of AR01 is primarily localized on the para-carbon atom of the pentafluorophenyl substitution group (see Figure 7a), while the LUMO of TPFPB is primarily localized on the boron center (see Figure 7e). Figure 7 clearly shows that the degree of LUMO localization on the boron center decreases in the following order: TPFPB (Figure 7e) > AR03 (Figure 7c) > TPB (Figure 7f) ~ AR04 (Figure 7d) > AR02 (Figure 7b) > AR01 (Figure 7a).

So far, this is the best correlation we have obtained between the theoretical data and the experimental electrochemical performance of lithium-ion cells, with reverse ordering of AR01 and AR02. We, therefore, believe that the molecular orbital localization is a key indicator for the electrochemical performance of the anion receptor. More theoretical and experimental cross-verification is needed for the applicability of this theoretical indicator to guide the search for a better anion receptor.

Conclusion

Four anion receptors were investigated as electrolyte additives for lithium-ion cells. The electrochemical results showed that the addition of tris(1,1,1,3,3,3-hexafluoroisopropyl) borate slightly improved the capacity retention and reduced the interfacial impedance. The electrochemical performance suffered greatly when 2-(2,4-difluorophenyl)tetrafluoro-1,3,2-benzodioxaborole or 2,5-bis(trifluoromethylphenyl)tetrafluoro-1,3,2-benzodioxaborole was used as the electrolyte additive. *Ab initio* calculations showed that the electrochemical performance had good correlation with the degree of localization of the lowest unoccupied molecular orbital of the anion receptors.

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